

Projected Reservoir Impairment in the Conterminous United States following Three Climate Change Scenarios

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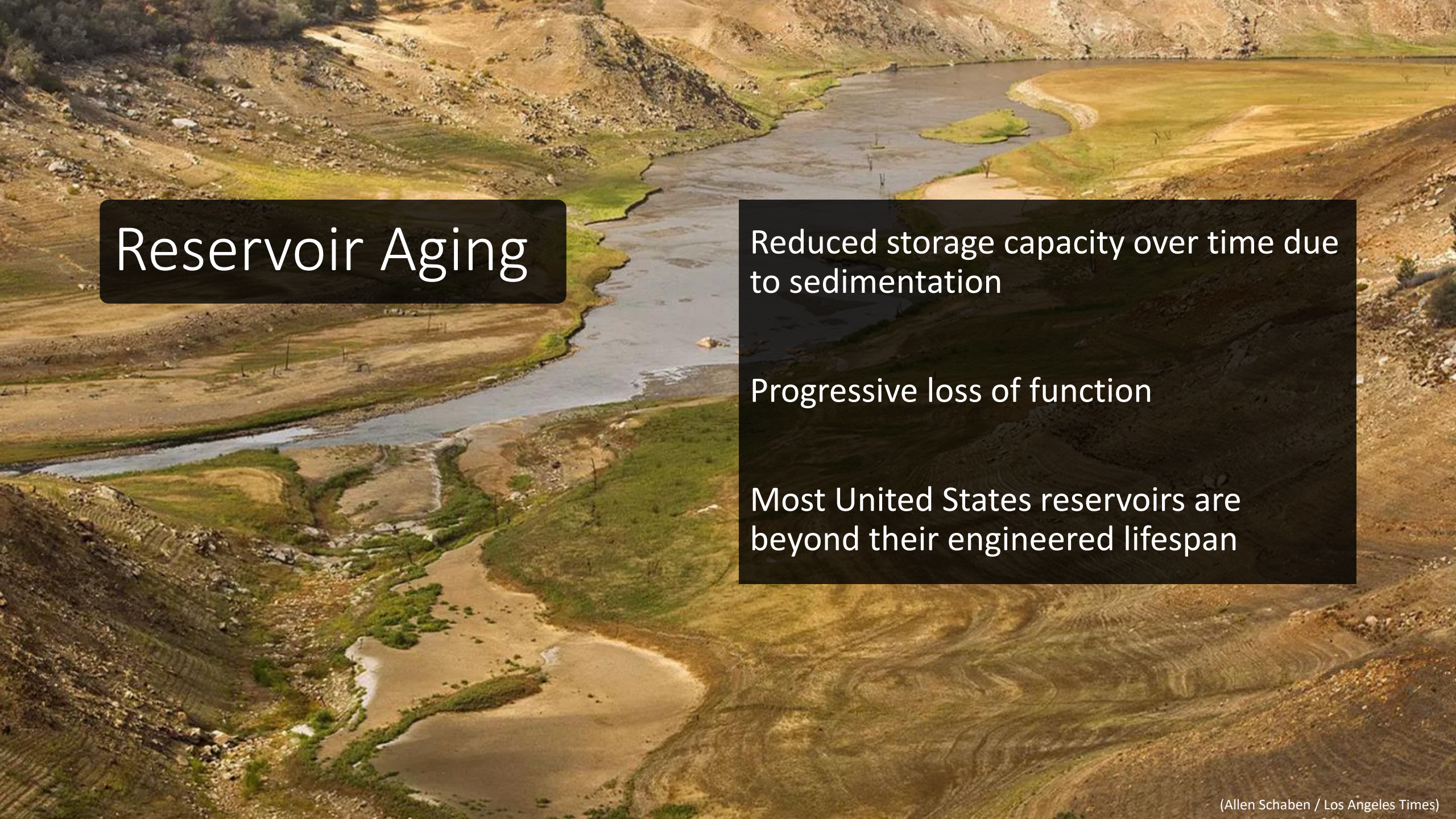


What is a Reservoir?







An aerial photograph of a reservoir showing extensive sedimentation. Large areas of dry, light-brown earth are exposed along the shoreline, indicating a significant drop in water levels. The water is a murky, greyish-brown color. The surrounding landscape is a mix of dry grass and rocky terrain.

Reservoir Aging

Reduced storage capacity over time due to sedimentation

Progressive loss of function

Most United States reservoirs are beyond their engineered lifespan

An aerial photograph of a winding river in a dry, hilly landscape. The river flows through a valley, surrounded by dry, brownish-green hills. The water is a light, silty color. The overall tone is warm and somewhat desaturated, suggesting a dry climate.

Climate Change Influences Aging

Extreme seasonality

Intense precipitation events

Longer droughts

Altered connectivity

Climate crisis

Climate crisis deepens with 2024 'certain' to be hottest year on record

Average global temperature in November was 1.62C above preindustrial levels, bringing average for the year to 1.60C

Damian Carrington
Environment editor

Sun 8 Dec 2024 22:00 EST

[Share](#)

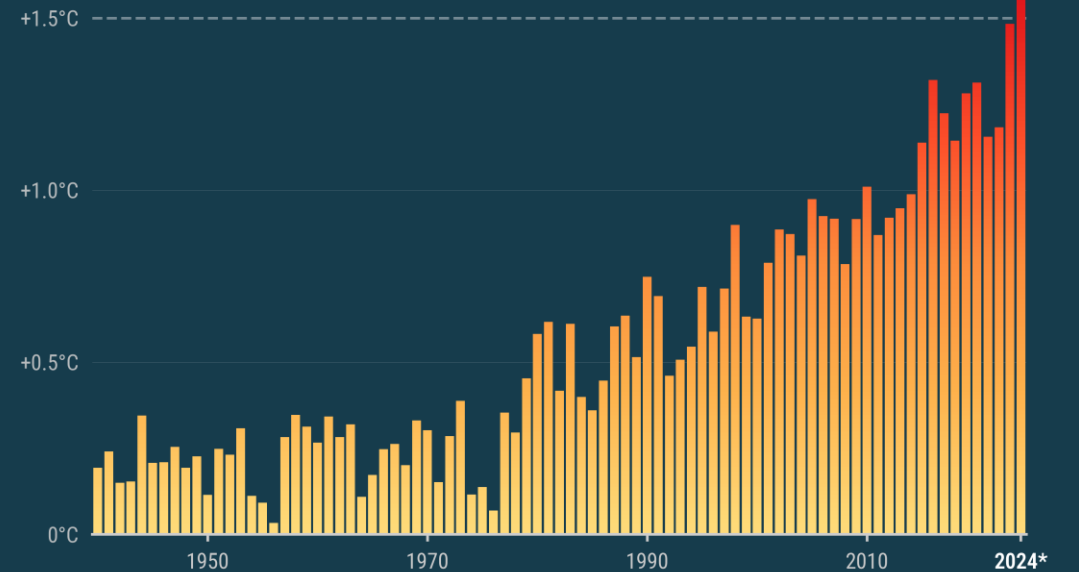


A wildfire in California this year. Fires driven by severe droughts have affected the western US, Canada, the Amazon forest and particularly the Pantanal wetlands. Photograph: David McNew/Getty Images

2024 on track to be warmest year and first year above 1.5°C

Annual global temperature anomalies relative to pre-industrial (1850–1900)

Data: ERA5 (1940–2024) • Credit: C3S/ECMWF



* Provisional estimate for 2024 based on 10 months (January to October)



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EUROPEAN UNION



IMPLEMENTED BY
ECMWF



Reservoir Managers and the Future



I need
electricity!

Save the turtles!

I'm thirsty!

I want to fish!



Objectives



Develop models to predict reservoir impairment from climate indicators



Identify managerially relevant groups of reservoirs facing similar climate conditions



Project reservoir impairment in the United States during the 21st century

Reservoir Impairment

Krogman and Miranda (2016), expanded from Miranda and Hunt (2010)

1090 Reservoirs

52 Impairment Metrics

Likert-type Ranked
Impairment Scores 0-5

Introduction

Broad view of reservoirs considering their watershed, tributaries, riparian zone, and tailwater

appropriate column):

Descriptions are available by hovering the pointer over each variable.

	None (0)	Low (1)	Low to Moderate (2)	Moderate to High (3)	Moderate to High (4)	High (5)
Excessively shallow reservoir	☐	☐	☐	☐	☐	☐

Current status of reservoir
(situation within last 5 years)

among reservoirs, answer in relation to similar reservoirs within your geographical region whether within your state or a nearby state. If there is another staff member in your agency that has more experience with this reservoir than you, please collaborate with that individual in formulating your responses. Because blank responses complicate analyses, a good guess is preferred over a blank response, but a blank response is better than a bad guess.

Google Earth

Data SIO, NOAA, U.S. Navy, NGA, GEBCO

Image Landsat / Copernicus

Data LDEO-Columbia, NSF NOAA

Image IBCAO

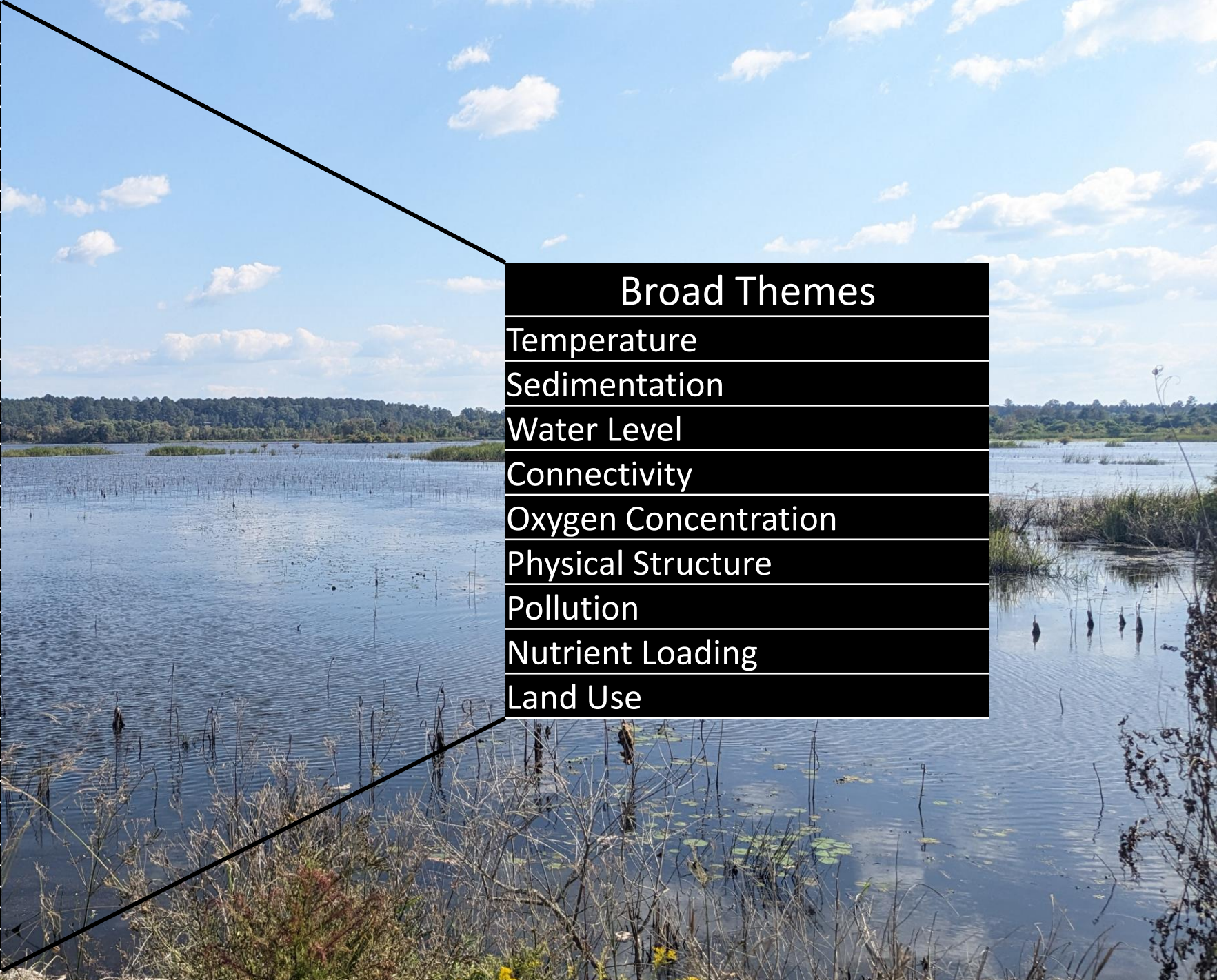
900 mi



Impairment Metric
Excessively shallow reservoir
Excessive littoral mudflats
Insufficient adjoining backwaters and wetlands
Insufficient connectivity to backwaters and wetlands
Insufficient connectivity to tributaries due to sedimentation
Excessive aquatic macrophytes
Insufficient aquatic macrophytes
Invasive plant species
Invasive animal species capable of altering habitat
Insufficient structural habitat
Excessively shallow littoral zone
Insufficient bank shading
Insufficient allochthonous inputs
Excessive disturbance of riparian zone
Harmful levels of agriculture in the surrounding watershed
Harmful levels of livestock production in the surrounding watershed
Harmful levels of logging in the surrounding watershed
Excessive nutrients
Insufficient nutrients
Excessive suspended sediments or inorganic turbidity
Excessive organic turbidity
Extreme seasonal variation in turbidity
Harmful algae blooms
Extreme diel variation in dissolved oxygen
Oxygen stratification
Excessively high temperatures
Excessively low temperatures
Temperature stratification
Untimely or frequent turnovers
Thermal pollution
Contaminants (heavy metals, biocides)
Non-point source pollution
Insufficient retention time
Insufficient water storage
Seasonally mistimed water level fluctuations
Excessive long-term drawdowns
Excessive short-term fluctuations
Rapid water level change
Sedimentation
Shoreline erosion
Loss of cove habitat due to depositional filling
Shoreline homogenization
Homogenization of littoral substrates
Disturbances in upstream watersheds
Disturbances in adjacent watersheds



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Broad Themes
Temperature
Sedimentation
Water Level
Connectivity
Oxygen Concentration
Physical Structure
Pollution
Nutrient Loading
Land Use

Climate Data

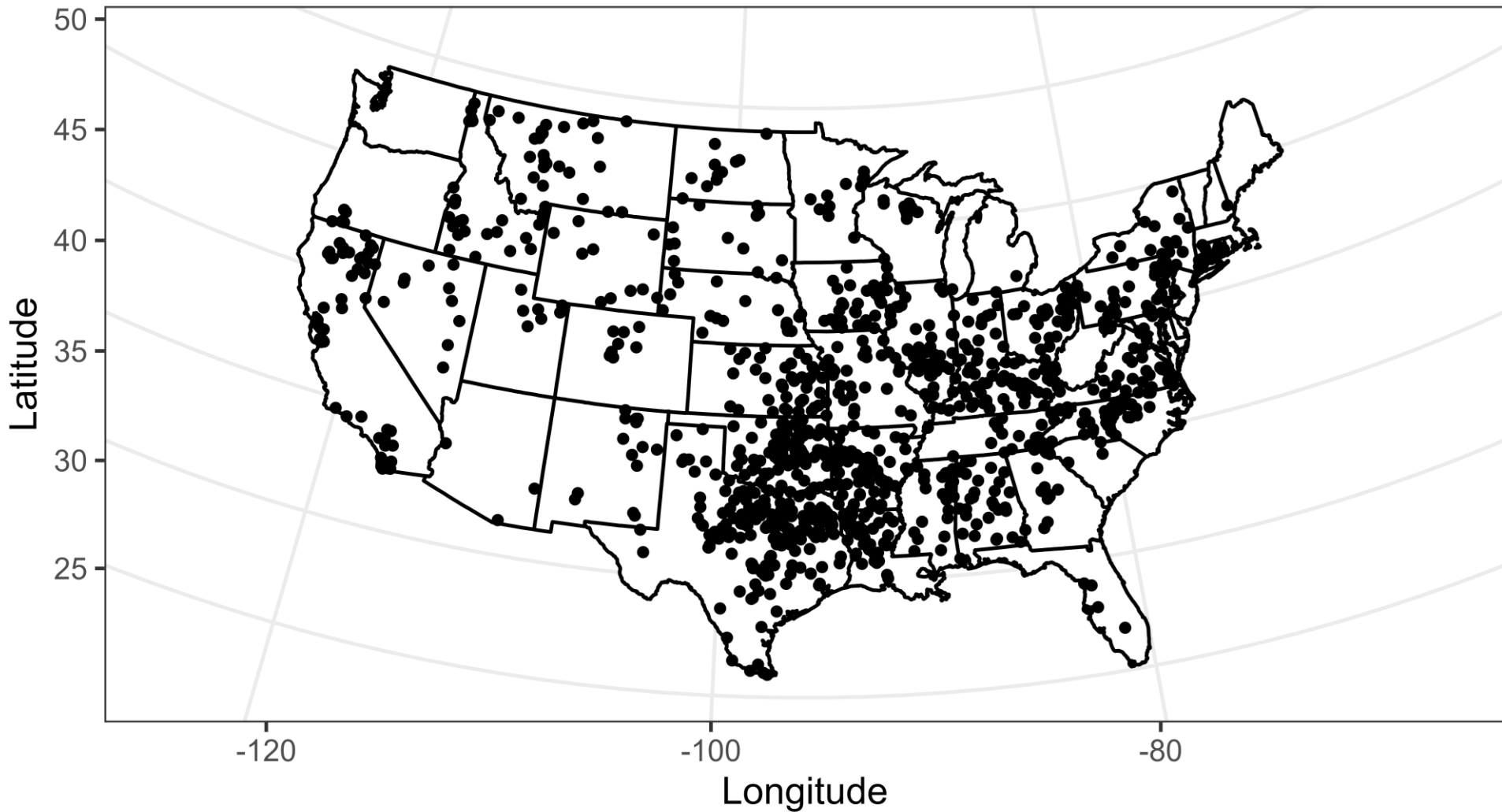
WorldClim Bioclimatic Indicators

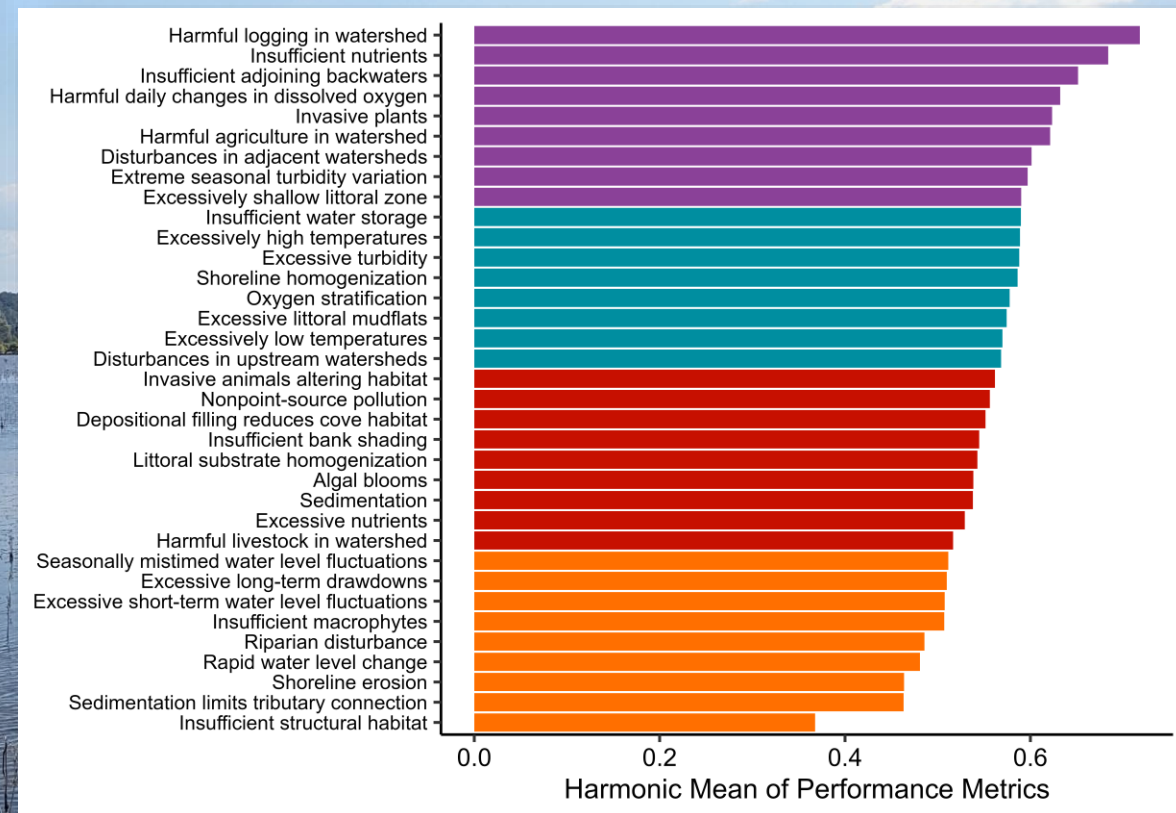
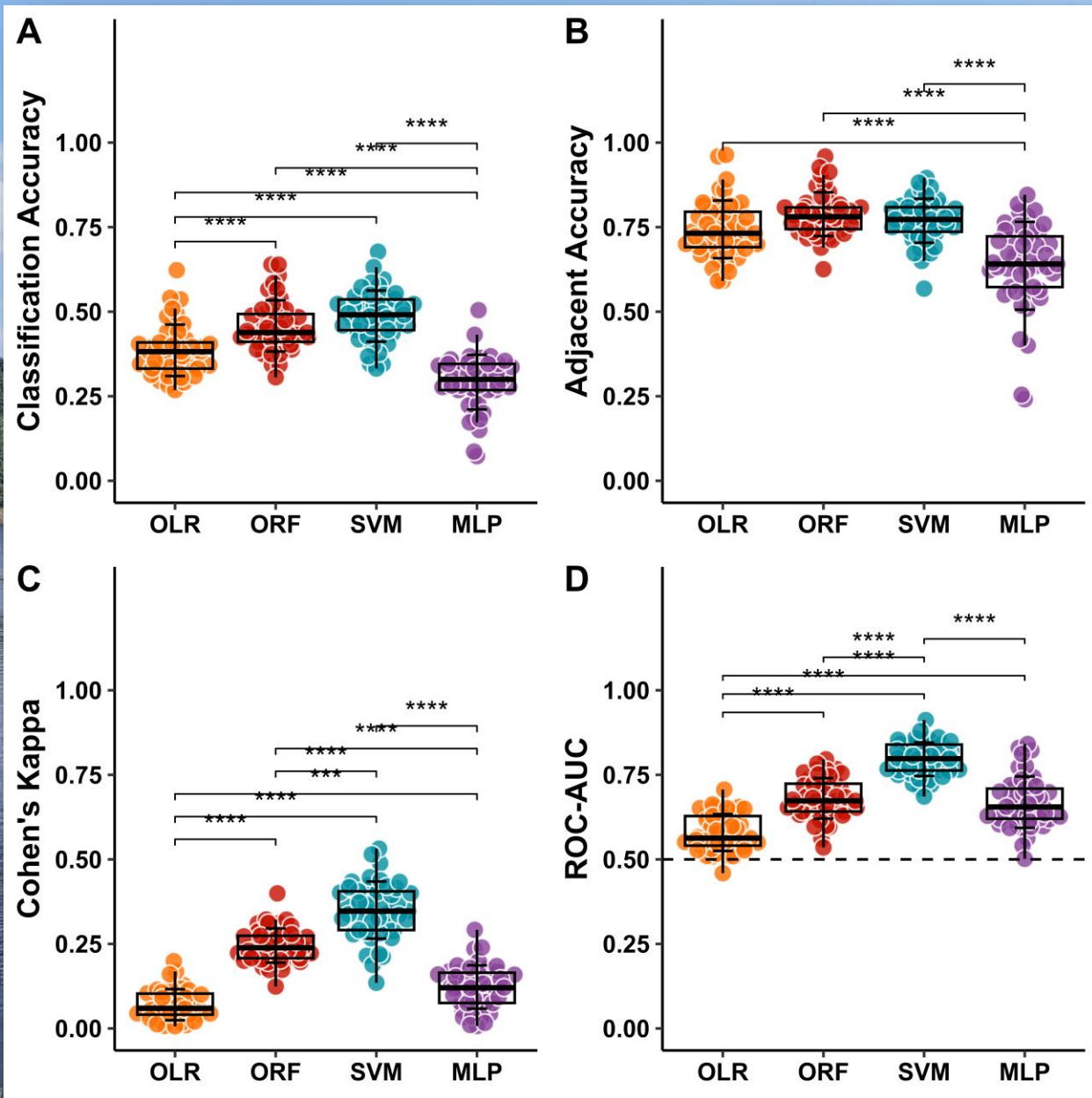
Annual Mean Temperature
Mean Diurnal Range
Isothermally
Temperature Seasonality
Max Temperature of Warmest Month
Min Temperature of Coldest Month
Temperature Annual Range
Mean Temperature of Wettest Quarter
Mean Temperature of Driest Quarter
Mean Temperature of Warmest Quarter
Mean Temperature of Coldest Quarter
Annual Precipitation
Precipitation of Wettest Month
Precipitation of Driest Month
Precipitation Seasonality
Precipitation of Wettest Quarter
Precipitation of Driest Quarter
Precipitation of Warmest Quarter
Precipitation of Coldest Quarter

Earth System Models

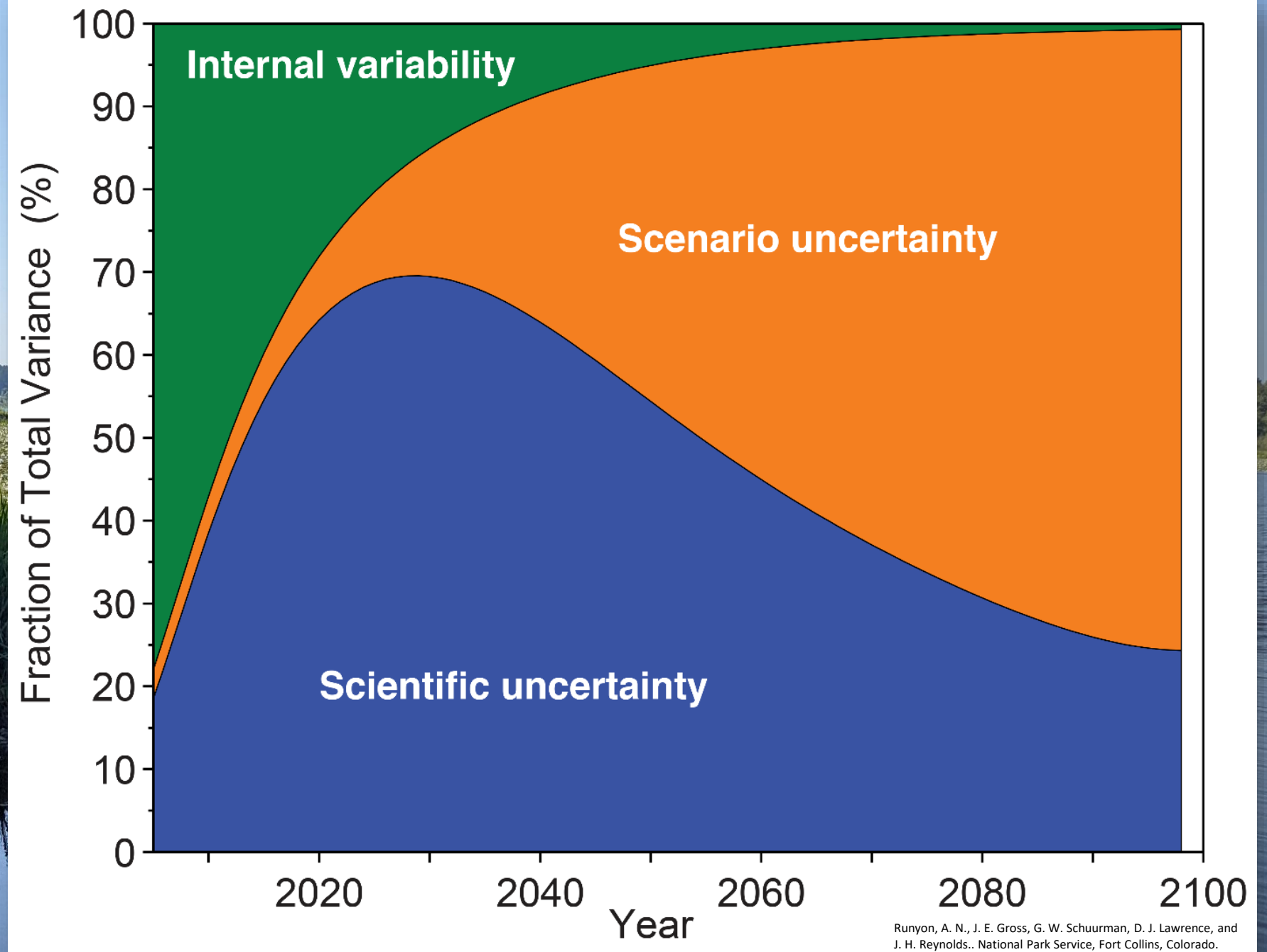
Model	Research Group
ACCESS-CM2	Australian Community Climate and Earth System Simulator
CMCC-ESM2	Fondazione Centro Euro-Mediterraneo sui Cambiamenti Climatici
EC-Earth3-Veg	EC-Earth3
GISS-E2-1-G	Goddard Institute for Space Studies
INM-CM5-0	Institute of Numerical Mathematics
UKESM1-0-LL	U.K. Earth System Model

Cluster Analysis

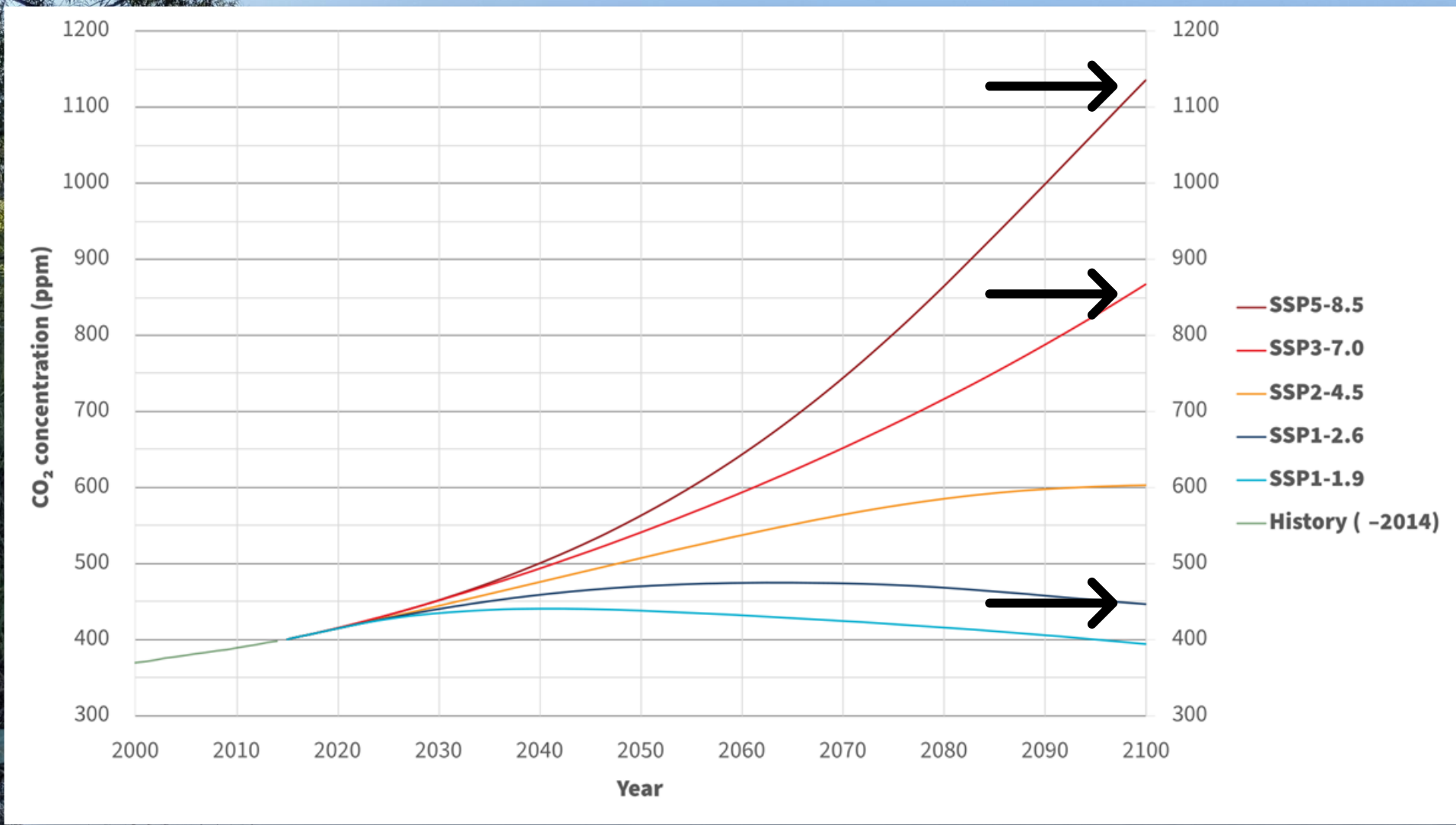




Uncertain Futures



The Shared Socioeconomic Pathways



Projected Reservoir Impairment

35 Impairment Metrics
1090 Reservoirs

Four Time Intervals

High Emissions

Moderate Emissions

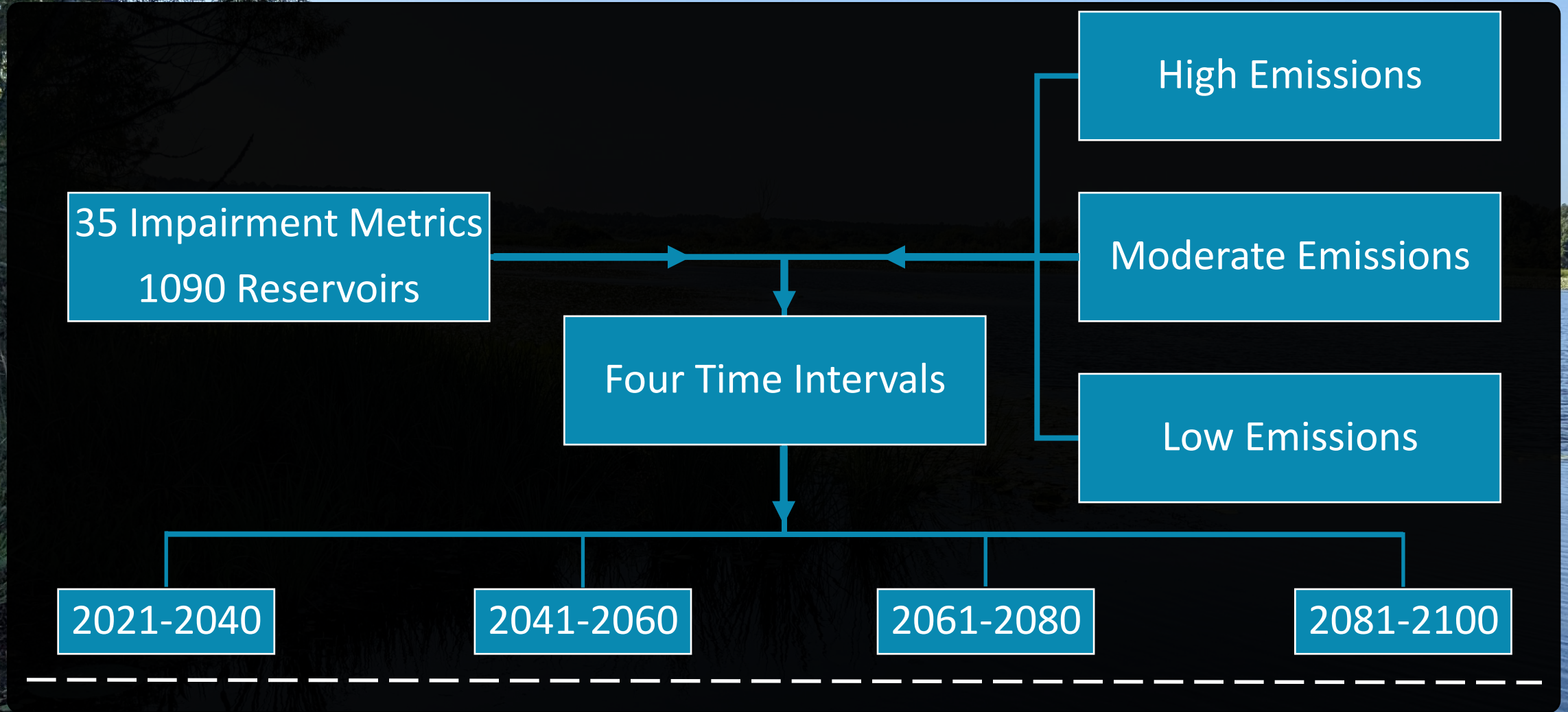
Low Emissions

2021-2040

2041-2060

2061-2080

2081-2100

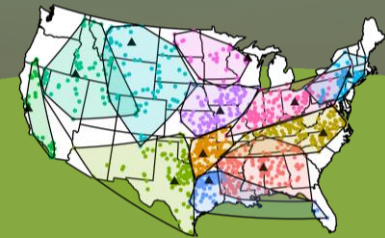


Spatial Extents of Results

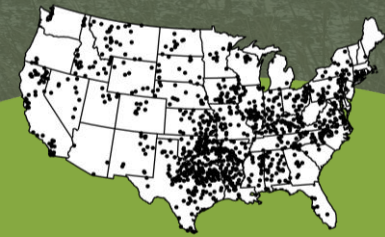
Conterminous United States



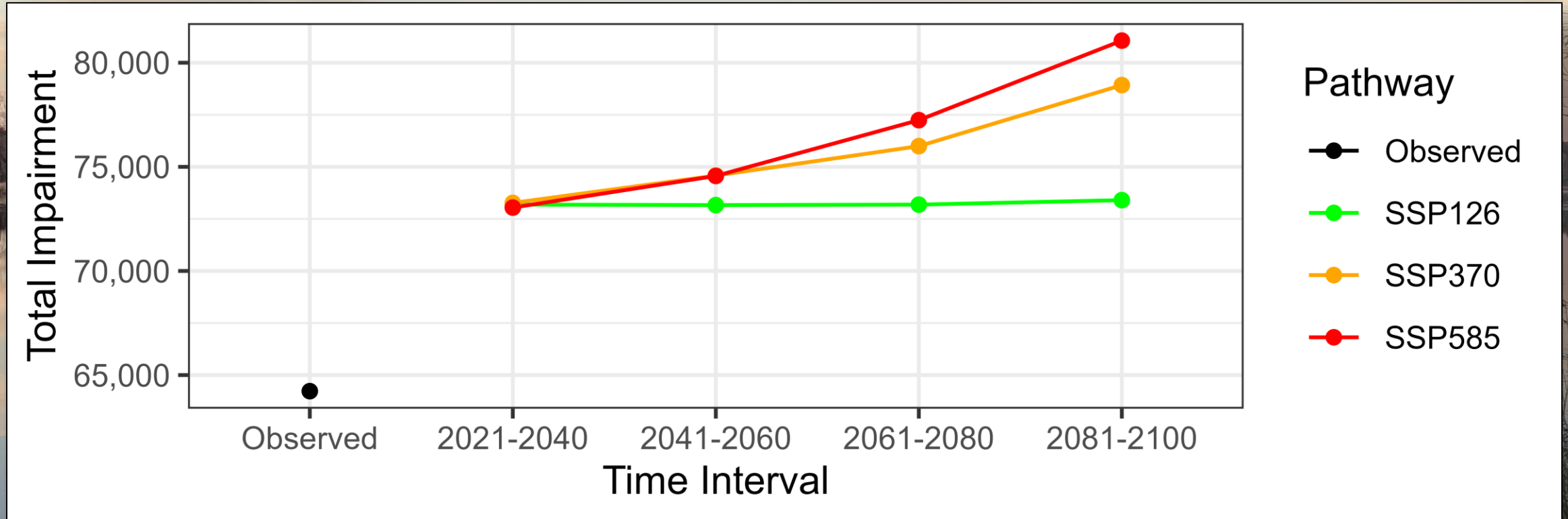
Climatic Cluster



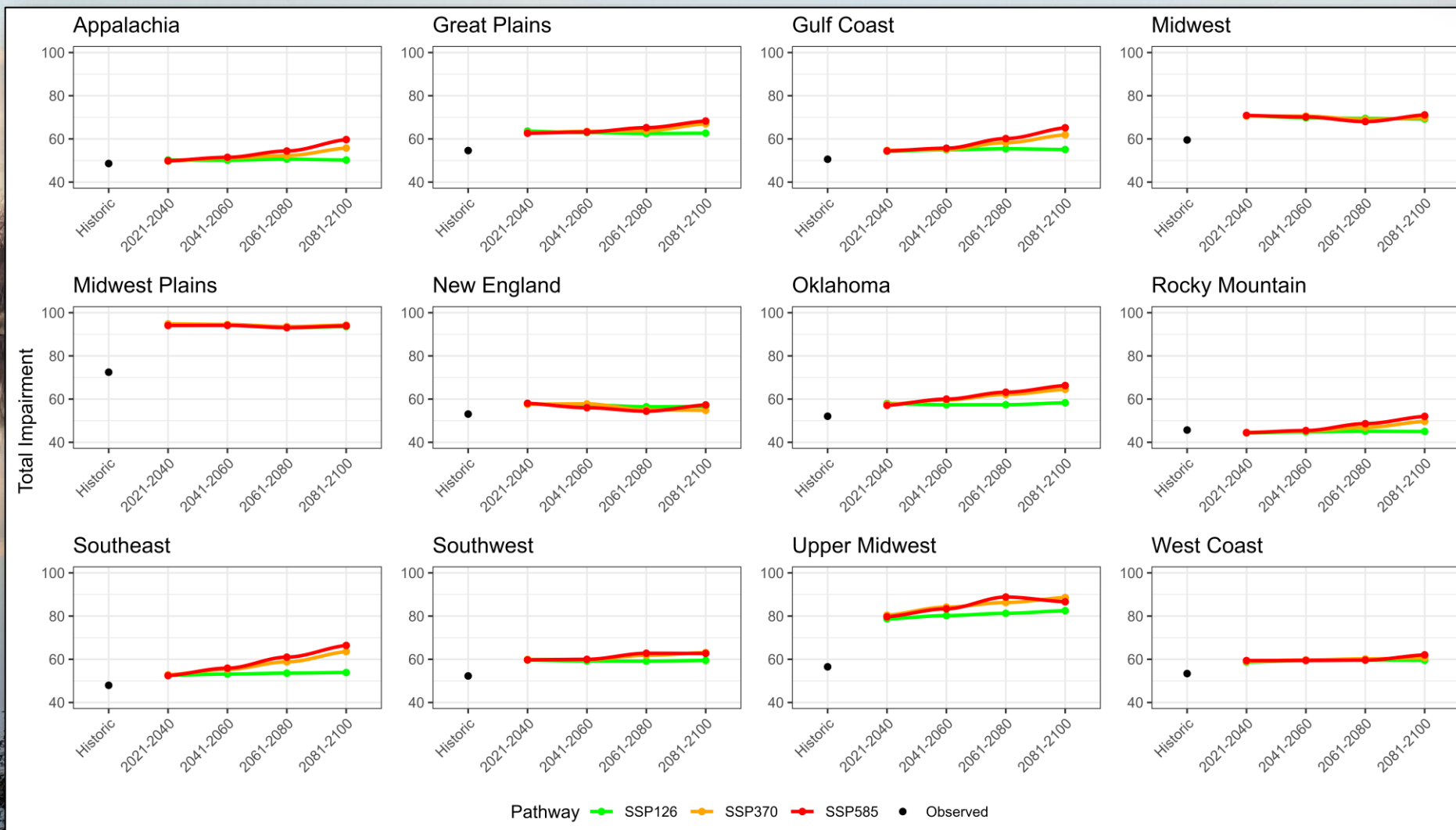
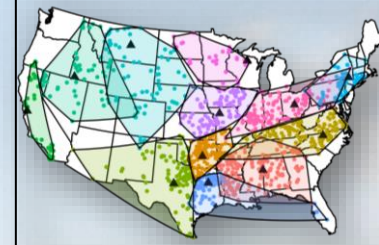
Individual Reservoir



Total Impairment for Conterminous United States



Total Impairment by Cluster



Trends Across the US:

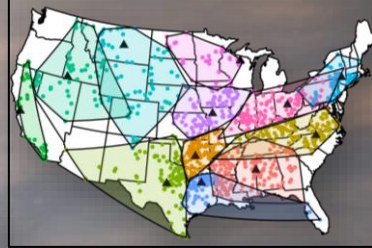
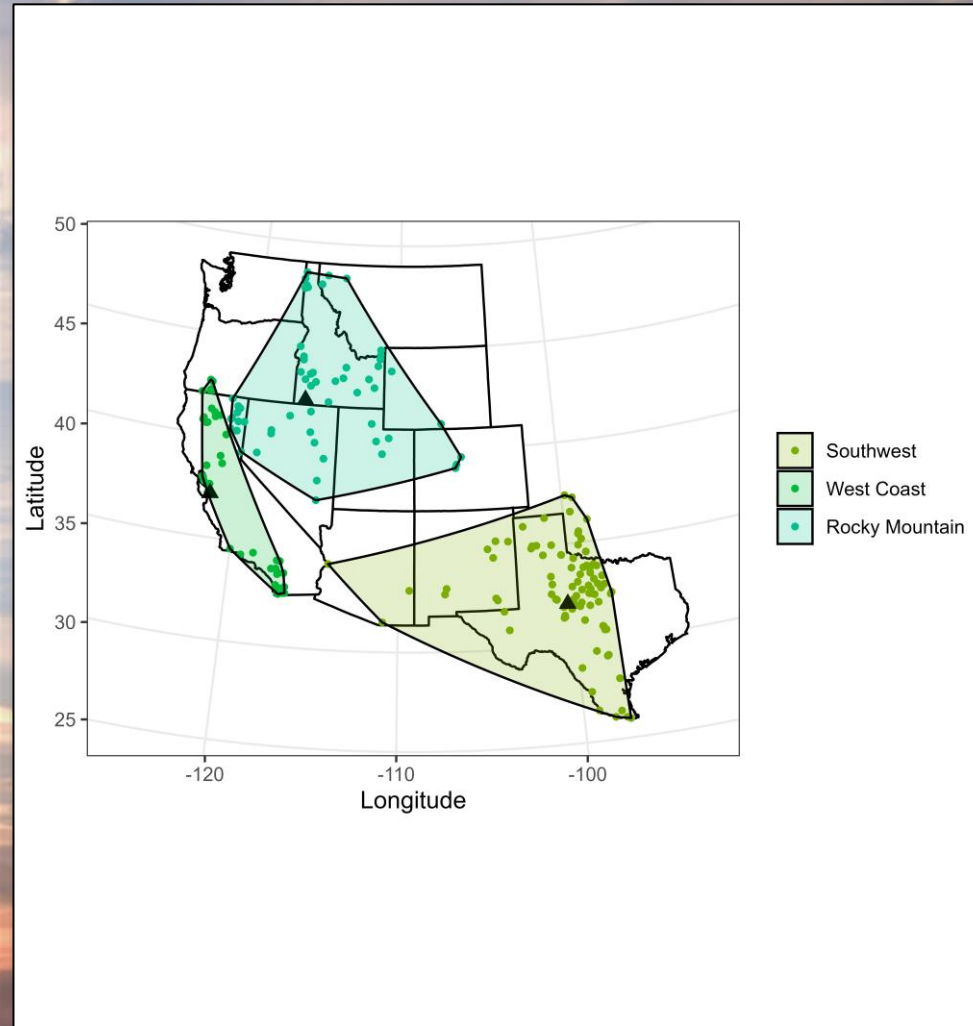
West Coast, Southwest, and
Rocky Mountains

Impairment generally associated
with water levels

Excessively shallow littoral zones

Excessive mudflats

Short-term water level
fluctuations

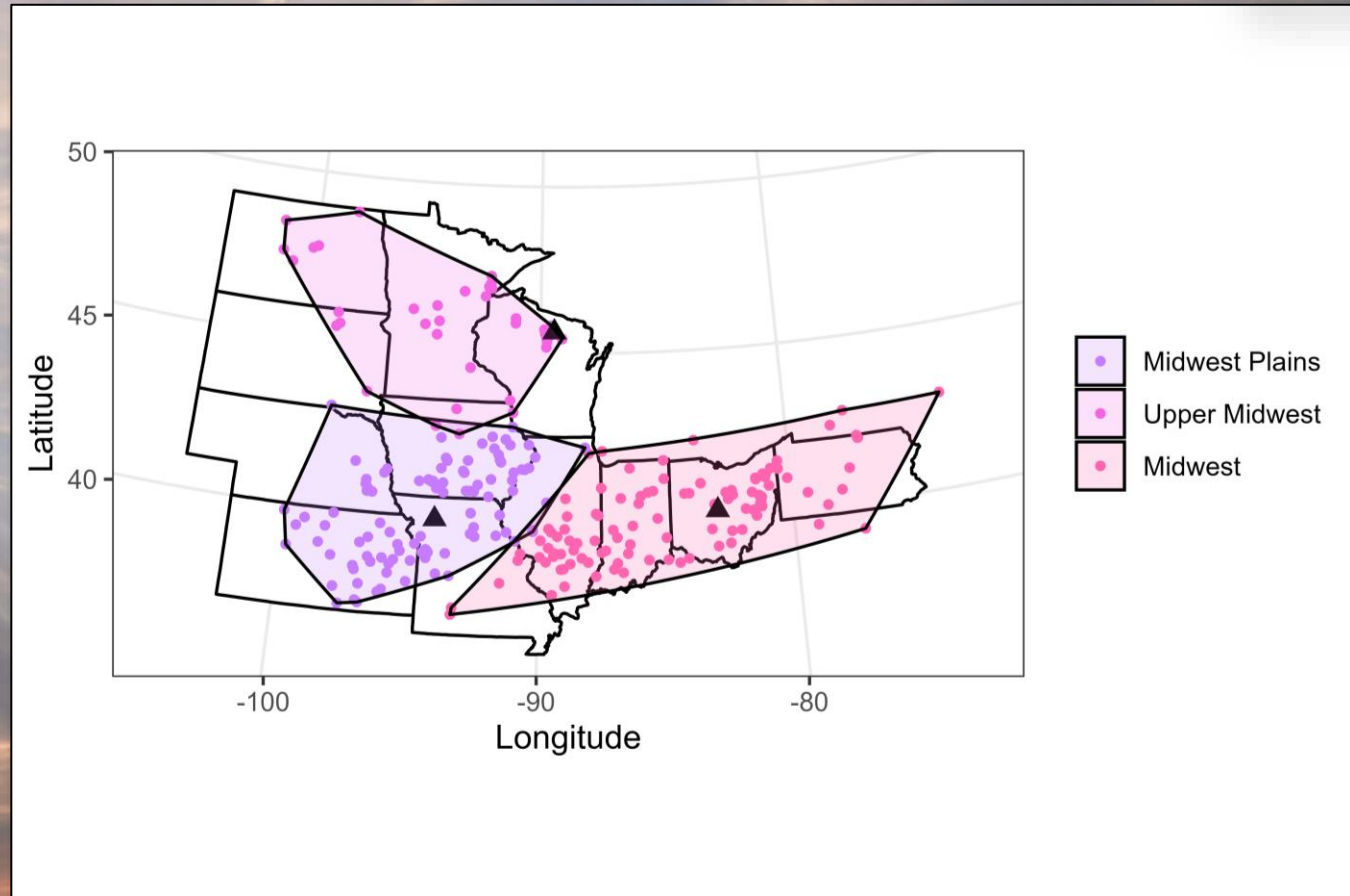
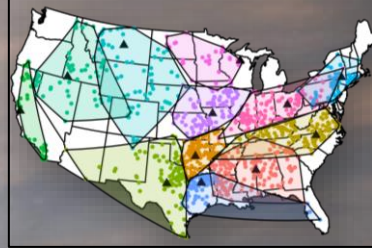


Trends Across the US:

Upper Midwest, Midwest,
and Midwestern Plains

Impairment generally associated
with land use

Excessive livestock and row-crop
agriculture in the surrounding
watershed



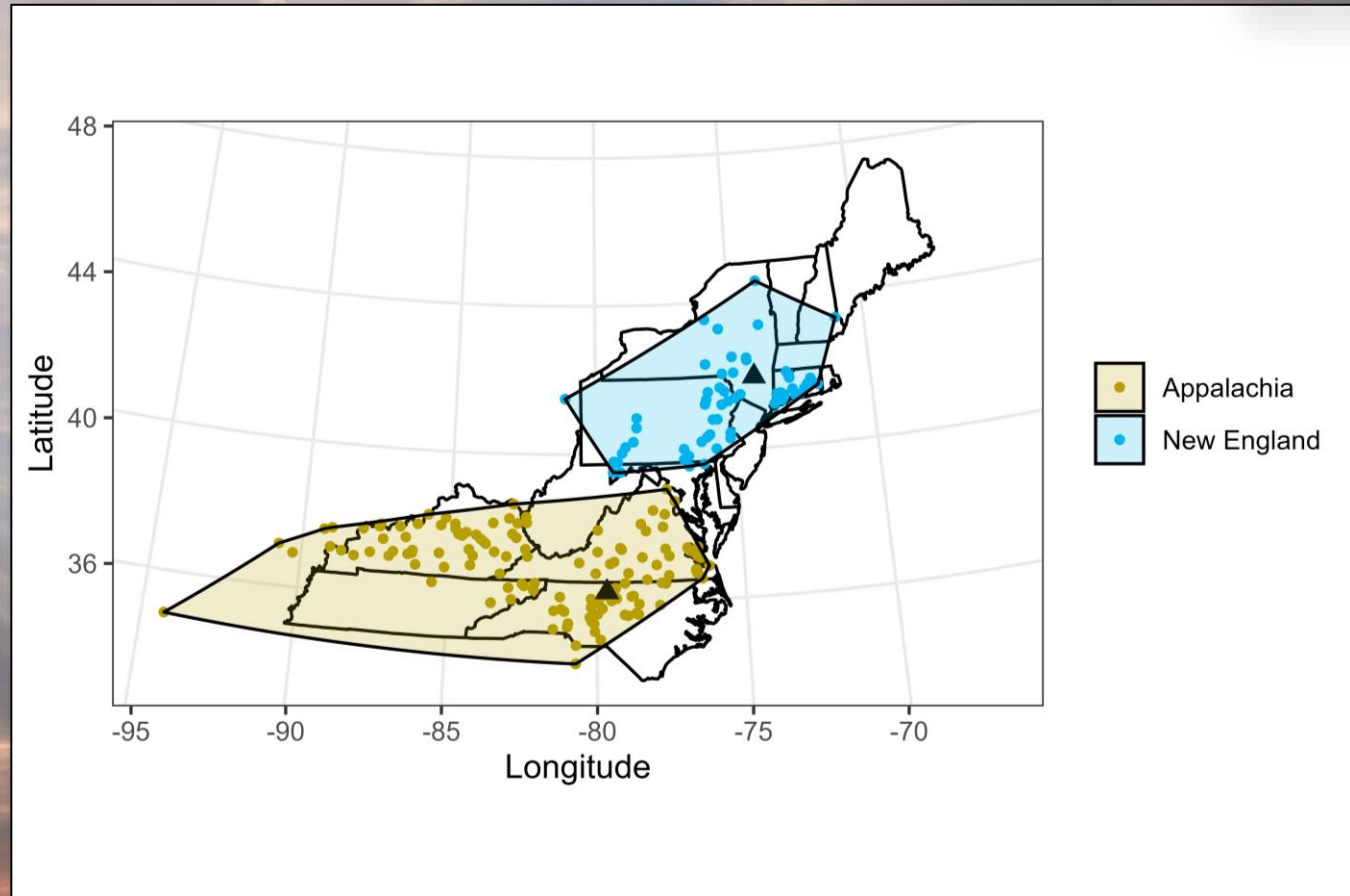
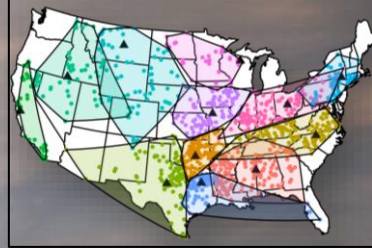
Trends Across the US: Appalachia and New England

Impairment generally associated
with shoreline degradation

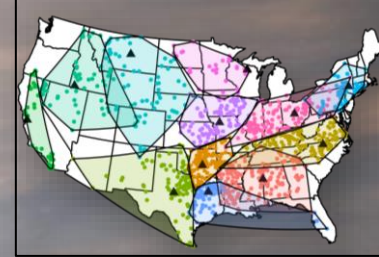
Sedimentation

Turbidity

Riparian Disturbance



Trends Across the US: Southeast and Gulf Coast

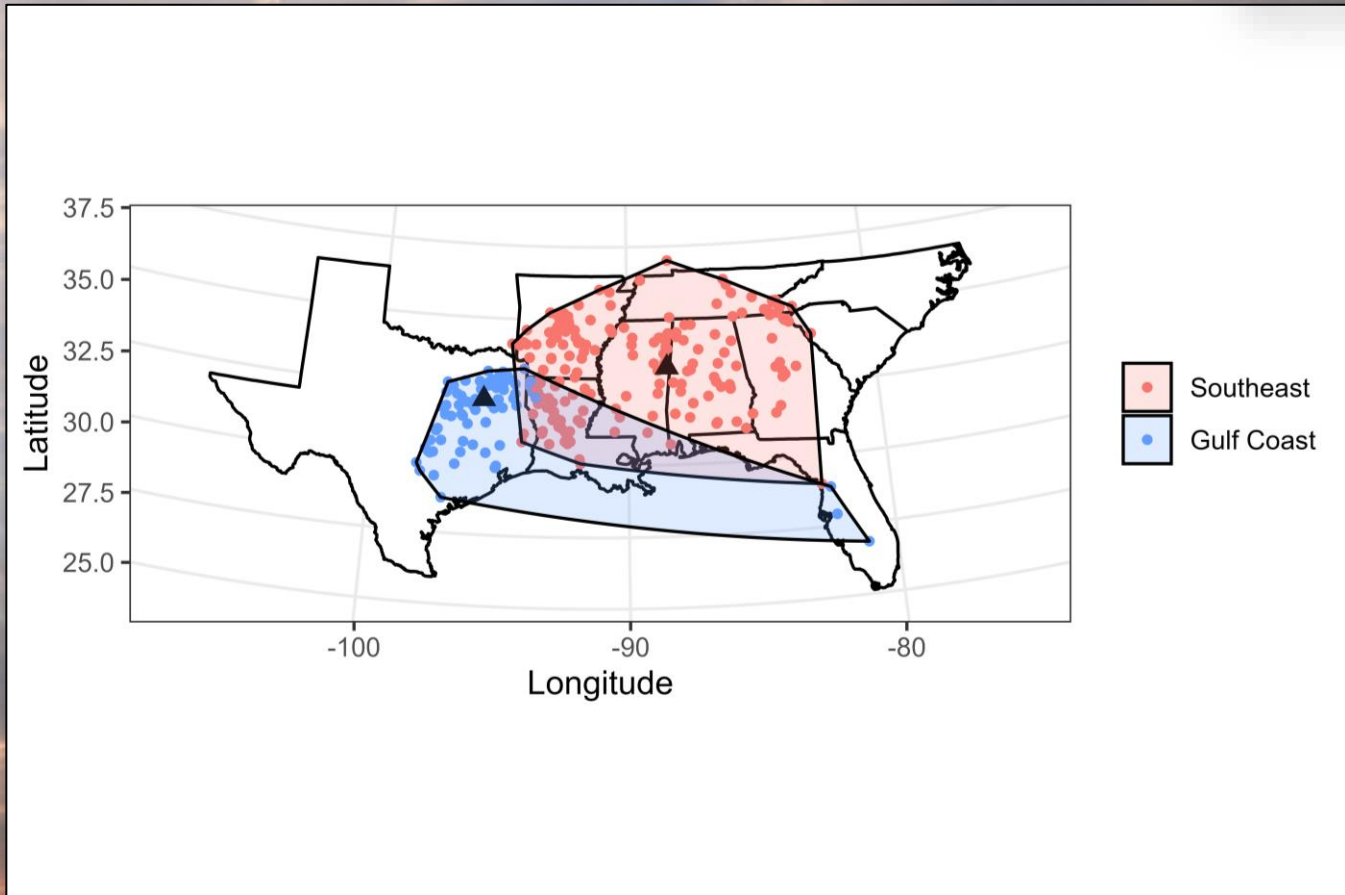


Impairment generally associated
with sedimentation

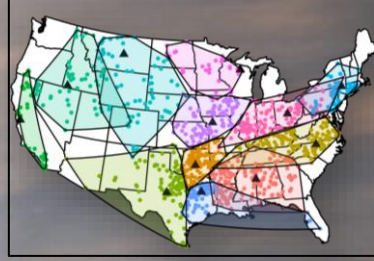
Sedimentation

Depositional filling

Reduced connectivity



Projected Impairment: Conclusions



- 1) Reservoir impairment varies by region and climate conditions
- 2) High impairment scores correspond with increased emissions
- 3) Similar impairment in moderate and high emissions scenarios
- 4) Managers should prepare for worst-case if possible

Res_Clim

Web application in R-Shiny for exploring reservoir impairment from climate change

Report describing impairment for reservoir and general projections for the relevant cluster

E:/R_Stuff/res_clim - Shiny
http://127.0.0.1:4404 Open in Browser Publish

Reservoir Impairment from Climate Change V0.1

Welcome to Res_Clim V0.1!

Res_Clim is a web application to explore reservoir impairment across the United States. The app presents a modeling effort to project reservoir impairment from climate change to inform management strategies and improve decision making. We leverage expert knowledge reported by reservoir managers to rank reservoir impairment on a 0-5 scale, model those ratings with bioclimatic indicators, and finally project future reservoir impairment following three climate scenarios.



All of the Impairment Metrics in Res_Clim

Impairment Metric	Definition
Excessive Littoral Mudflats	Seasonally flooded and exposed expansive layers of soft sediments; terrestrial vegetation seldom grows unless the mudflats are exposed for many months.
Insufficient Adjoining Backwaters	The reservoir or tributaries have no or limited adjoining backwaters or wetlands and therefore lack the benefits of those habitats.
Sedimentation Limits Tributary Connection	Sedimentation has resulted in decreased connectivity to tributaries during low-flow periods, acting as a barrier to fish movement.
Insufficient Macrophytes	Lacking or deficient aquatic plants for structural fish habitat.
Invasive Plants	Presence of nonnative aquatic macrophytes that may negatively impact reservoir systems, reduce public access, or present other problems to reservoir managers.



Filter Results by User Needs



Reservoir Impairment from Climate Change V0.1

Home Study Reservoirs Additional Reservoirs About GitHub

Reservoirs in Study

This page provides projected impairment for over 1000 reservoirs! If you know the name or National Inventory of Dams ID (NIDID) of the reservoir you're interested in, try the search menus below. Otherwise, you can pan the map to find your reservoir.

Filter by State? ☐

Reservoir Name

NIDID ?

Select Impairment Metric(s) ?

Select Time Intervals(s) ?

Select Climate Pathway ?

Submit

Reset All

Status: Server Ready

Lat: 32.5890094863584
Lng: -89.2638347925361

Downloadable Report

Impairment Report for OKTIBBEHA COUNTY LAKE DAM

ChardarYGO
2025-01-19 18:36:22

Shared Socioeconomic Pathways

The shared socioeconomic pathways (SSPs) describe potential future narratives for society which could lead to a range of potential economic futures. In general, reservoirs are projected to face more severe impairment following higher emissions scenarios.

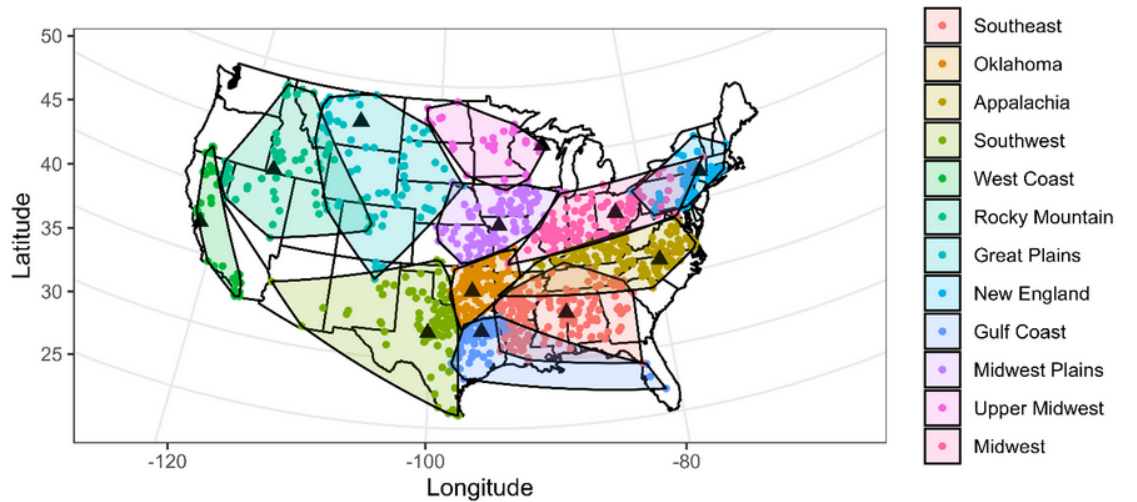
You have selected SSP585. SSP585 is a high greenhouse gas emissions scenario characterized by strong economic growth through intense exploitation of fossil fuel resources, with optimistic potential for future technological development to reduce environmental impacts.

Table of Projected Impairment for OKTIBBEHA COUNTY LAKE DAM

This table provides the projected impairment scores for each metric you selected.

##	Year	Pathway	Metric	Score
## 1	2081-2100	SSP585	Excessive Littoral Mudflats	5
## 2	2081-2100	SSP585	Insufficient Structural Habitat	1
## 3	2081-2100	SSP585	Riparian Disturbance	4
## 4	2081-2100	SSP585	Sedimentation	5

Climatic Cluster



Reservoirs across the conterminous United States were grouped into climatic clusters based on similar climate conditions. OKTIBBEHA COUNTY LAKE DAM is in the Southeast cluster. The Southeast cluster is characterized by high impairment related to sedimentation, including depositional filling and reduced connectivity. This may be driven by high precipitation increasing discharge and sediment load into reservoirs in this region.



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Katie Baker



Questions?

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